ICS 6B F26 Take Home Exam 4

Due: October 23, 2026 at 11:59PM

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UCI NetID :

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(**alpha-numeric; NOT your student ID**)

- Read the instructions of each question carefully.
- Show your work. Where applicable, partial credit will be given on the work shown.
- Please keep in mind the academic honesty guidelines. This take-home exam is to be completed individually, with no outside help. You may use any resources from our class (ZyBooks and resources from Canvas), but you may not use any other online resources.
- You may choose to print the exam or use a digital editor for completing the exam. It is required that you use this PDF to complete your work. If you have no access to a printer or digital tools to fulfill the exam, feel free to reach out to the staffs regarding your concerns.
- If you have any questions, please post a private Ed or attend available Office Hours. Note that we are not allowed to provide specific help to answering the exam questions.
- There are **4** problems on this test.
- There are a total **16** of points.
- Good luck!

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Problem 1 (3 points)

Consider the sets A and B defined as follows:

$$A = \{x \in \mathbb{Z} : \ (x < x^2) \ \wedge \ (x > x^3)\}$$

$$B = \{x \in \mathbb{Z} : \ \exists k \in \mathbb{Z} \ (x = -2k)\}$$

1. Choose **all** options that are **elements** of $A \times B$.
   ☐ $(-2, -1)$ ☐ $(-1, -2)$ ☑ $(-2, -2)$ ☑ $(-2, 2)$ ☑ $(-2, 4)$ ☐ $(4, -2)$
2. Choose **all** options that are **elements** of $B \times A$.
   ☐ $(-2, -1)$ ☐ $(-1, -2)$ ☑ $(-2, -2)$ ☐ $(-2, 2)$ ☐ $(-2, 4)$ ☑ $(4, -2)$
3. Choose **all** options that are **elements** of $(A \times B) \oplus (B \times A)$.
   ☐ $(-2, -1)$ ☐ $(-1, -2)$ ☐ $(-2, -2)$ ☑ $(-2, 2)$ ☑ $(-2, 4)$ ☑ $(4, -2)$

Problem 2 (3 points)

We define sets A and B to be:

$$A = \{x \in \mathbb{Z} : x \text{ is one more than a multiple of } 6\}$$

$$B = \{x \in \mathbb{Z} : x \text{ is a prime number.}\}$$

1. $S_1 = A \cap B$. Choose **all** the **elements** that belong to the **power set** of S_1.
   ☑ $\varnothing$ ☑ $\{13\}$ ☐ 7 ☐ $\{3\}$ ☐ $\{-5, 1\}$ ☐ $\{-11, 2, 9\}$
2. $S_2 = A \cup B$. Choose **all** options that are **subsets** of the **power set** of S_2.
   ☑ $\varnothing$ ☑ $\{\varnothing\}$ ☐ $\{5, 7\}$ ☑ $\{\{1\}, \varnothing\}$ ☐ $\{\{-5, 1\}, \{\varnothing\}\}$ ☑ $\{\{-11, 1, 25\}\}$
3. $S_3 = B \times \{\varnothing, \{\varnothing\}\}$. Choose **all** the **elements** that belong to the **power set** of S_3.
   ☑ $\varnothing$ ☐ $(3, \{\varnothing\})$ ☐ $\{(\varnothing, 5)\}$ ☑ $\{(2, \{\varnothing\}), (3, \varnothing)\}$ ☐ $\{\varnothing, (7, \{\varnothing\})\}$ ☐ $\{\varnothing, \{\varnothing\}\}$

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Problem 3 (4 points)

Consider a baseball game with an attendance of 50,000. Determine whether the given subsets form partitions for the 50,000 attendees or not. Note that for a sequence of sets $B_1, B_2, \ldots, B_n$ to be a partition of B, they need to satisfy the following 4 properties:

- Property 1: $\forall i, B_i \subseteq B$
- Property 2: $\forall i, B_i \neq \emptyset$
- Property 3: $B_1, B_2, \ldots, B_n$ are pairwise disjoint i.e. $\forall i, j \ (i \neq j) \rightarrow (B_i \cap B_j = \varnothing)$
- Property 4: $B = \bigcup_{i=1}^{n} B_i$

For each part, choose the corresponding properties that are satisfied. Briefly explain why each property is/isn't fulfilled in the specified box.

a) There are k different places that serve food in the stadium. Any attendee who eats at one of these places is counted in the subset of attendees who have eaten there. Attendees can eat from multiple different places. Some of the attendees do not eat any food and some of the places that serve food do not get any attendees.

☐ Property 1 ☐ Property 2 ☐ Property 3 ☐ Property 4 ☐ It is a partition.

Explanation:

1. All food stations only get guest from people who attending the game

2. Some stations didn't get attendees, so there's empty set

3. Some attendees went to multiple stations, so one element appears in different sets

4. Some of the attendees didn't eat any food, so the stations doesn't have all elements of B

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b) There are m different sections in the stadium. Every section has multiple attendees and no attendee has tickets to multiple sections. All attendees have tickets to some section. A subset is defined for each section by the attendees who have tickets for that section.
☑ Property 1 ☐ Property 2 ☑ Property 3 ☐ Property 4 ☑ It is a partition.
Explanation:

1. Only attendees are accessing the stadium sections

2. All sections has multiple attendees, so no empty section

3. No attendees have tickets to multiple sections, so pairwise disjoint

4. All attendees have tickets to some section, so it covers all attendees in B

Problem 4 (6 points)

Consider the sets A and B such that $|A| = n$, $|B| = m$, and $|A \cap B| = n - 1$. Additionally, $n = m + 1$. For each of the following questions write your final answer in the box and show your work in the bigger box that follows. If the solution cannot be determined from the given information then write *'unknown'* and briefly explain why.

1. $|A \cup B| =$ n

Explanation:

Since n = m + 1, we have m = n − 1
thus |A∩B|=n−1=m=|B|
since (A∩B)⊆B, and they have the same size, thus B⊆A
thus A∪B=A. Thus |A∪B|=|A|=n.

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2. $|P(A) - P(B)| =$ 2^m

Explanation:

From question 1, we know B⊆A and A has exactly one more element than B.
Therefore P(B)⊆P(A)
Thus there are 2^n - 2^m = 2^(m+1) - 2^m = 2^m

3. $|P(A) \cup P(B)| =$ 2^n

Explanation:

From question 2, we know P(B)⊆P(A)
thus P(A)∪P(B)=P(A)
Thus |P(A)|=2^n

4. $|P(P(A \cap B))| =$ 2^(2^m)

Explanation:

From question 1, we know |A∩B|=|B|=m
Thus |P(P(A∩B))|=|P(P(B))|=2^(2^m))

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5. $|A^2 - B^2| =$ 2n - 1

Explanation:

From question 2 we know B⊆A
Thus B^2 ⊆ A^2
Thus |A^2−B^2|=|A^2|−|B^2|=n^2−m^2 = n^2 - (n-1)^2 = 2n - 1

6. $|B^2 \times A^2| =$ n^2 · m^2

Explanation:

From question 5 we know A^2 =A×A and B^2 = B × B
And |A^2|=n^2,|B^2|=m^2
Thus |B^2×A^2|=|B^2|·|A^2|=m^2 · n^2